Exploring Nature's Cooling Effect Through a Digital Twin

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1. Case description

1.1. Personal Motivation

In the modern digital age, the knowledge of the importance of nature is degrading quickly. Especially, children don't always realize that the importance of the environment can impact the area around them, but also the whole ecological system. In times where quality of information tends to vary, we want to provide kids with a fun, interactive tool to show how vegetation impacts temperature. The goal is to show how green spaces and forests impact their daily lives and to make them appreciate nature more.

1.2. Description of Groenlo

Reinrinck is a hostel near Groenlo in Oost Gelre, the Netherlands. This area has a diverse landscape, from small towns, agricultural grasslands, and a few forestry patches. This area lies in the municipality of Oost Gelre, which consists of 29.868 inhabitants (allecijfers.nl, 2026). This area can be considered a region with population decline, where most people in the region are between the ages of 54 and 65. The forest patches we chose to visualize are close to the hostel we were staying at. This was partially due to convenience and time limit, but also the exact location is less important to visualize the effect we are interested in.

Figure 1. Aerial photography of the Reinrinck shows the plots we chose to visualize using Lidar and thermal imaging.

To make the application more realistic, we chose to recreate the Reinrinck area with these patches of detailed forest, to some extent.

1.3. Target User Group

The target user group includes individuals aged 12 to 14, who just started high school they will have a basic understanding of technology and will be able to navigate the game. Drawing on terrestrial laser scanning (TLS) data collected in the Groenlo area, we aim to develop an immersive Unity-based game that allows users to explore and interact with a virtual forest environment. This user group has been chosen so they learn about the importance of being surrounded by vegetation and what impact it has at a young age.
This target audience will likely have some base knowledge of the cooling effect of shade. By showing the impact that trees have on their surroundings with a fun interactive gaming experience this bridges the knowledge gap.

1.4. Unity game description.

1.4.1. Overview

'Exploring Nature's Cooling Effect Through a Digital Twin' is a virtual Unity-based experience. Players navigate freely through a three-dimensional forest environment, where real TLS point cloud data is embedded within a richly textured scene featuring natural vegetation, terrain, and an ambient soundscape. As users explore the environment, they can interact with individual trees to discover key ecological concepts such as canopy height communicated through intuitive, color-coded visualizations that require no prior technical expertise. For later development, more interactive information could be given when interacting with a tree. This first-person game aims to interact with field data while you move around inside a 3D scenario where the player can move around, visualize the landscape, and obtain information from it. The virtual experience will have ambient music such as water, birds and wind to make a more "real" experience.

1.4.2. Functional Requirements

- Create a 3D immerse game scenario for forest landscapes.
- TLS; take measurements from specific ground sites considering parcels of 15x15 meters.
- Temperature continuous map using drone thermal camera imagery. This will be place as ground texture inside Unity.

1.4.3. Tools

- VZ-400 to make a LiDAR scan. This is used to measure local biomass and create a framework for unity trees.
- Thermotracer TH9100 to get an aerial thermal scan
- Unity software.

1.4.4. Data Preparation and Processing

The project will be divided in two parts. The initial LiDAR and thermal imagery data collection in Groenlo and the second part focus on the development of the immerse scenario on unity.

1. The TLS data will be collected in plots of 15 by 15 meters. Considering each plot, all vegetation contained inside the plot will be used for later processing.
2. For drone thermal imagery, we will use the delimited polygon over the forest area to create the flight path considering a side overlapping of at least 75%, a constant flight height by using a ground digital elevation model (to avoid GSD distortion in different images with different elevations) and a maximum flight velocity of 5m/s to ensure good quality on the images and avoid the blurry effects on the final mosaic.
3. TLS data will be uploaded into Cloud Compare software to perform tree segmentation, ground points removal and subsample selection. Because Unity has its own render limit capacity, heavy files should be treated carefully.
4. For the drone imagery, the Agisoft metashape software will be used to process the drone images and apply all the calibrations necessary for thermal imagery. After the initial quality assessment, the mosaic tool will be used to create the Ortho mosaic for the temperature map over the area.
5. Unity version 2021.03.45 will be used to create the 3D immerse scenario. The final product will be the unity project with the link access to the visualization (only if this is possible). Inside the game, the objective is that a person can walk around the forest and interact with specific trees to obtain different information described above. As a prototype application, basic functionality will be present (instead of more complex interaction with the surrounding environment. See Figure 2 and 3 for expected results.

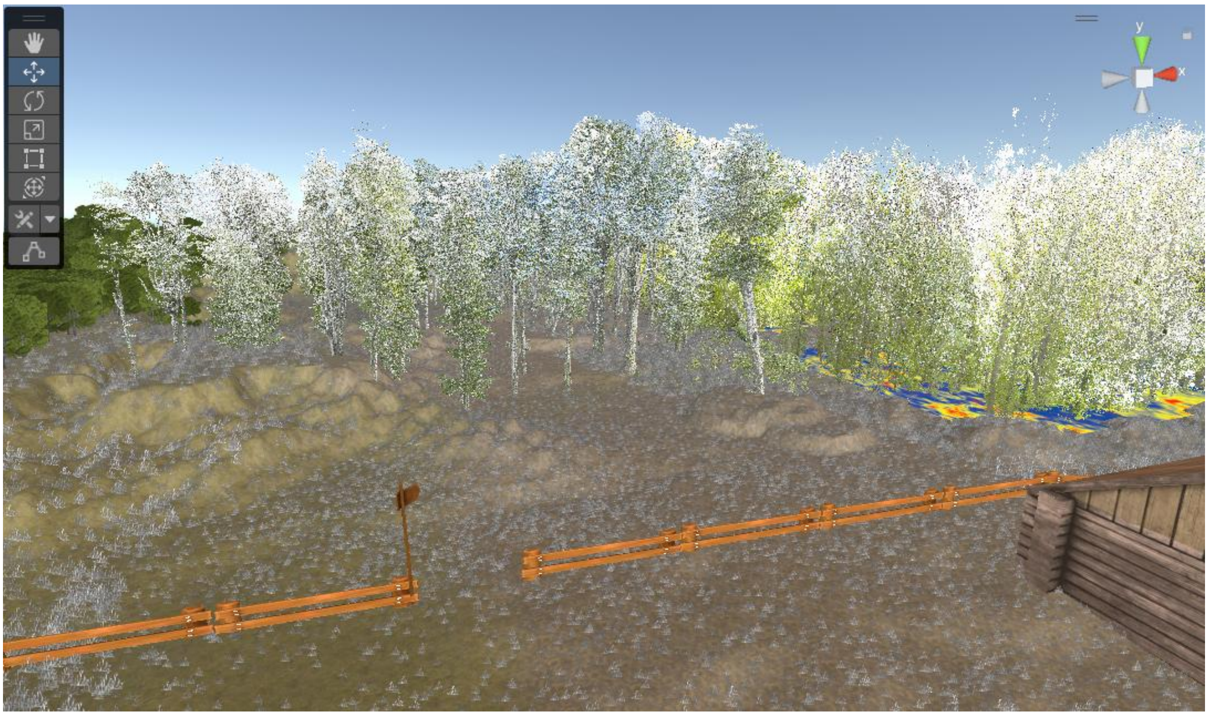


Figure 2: A patch of forest in our application, through the player's POV.

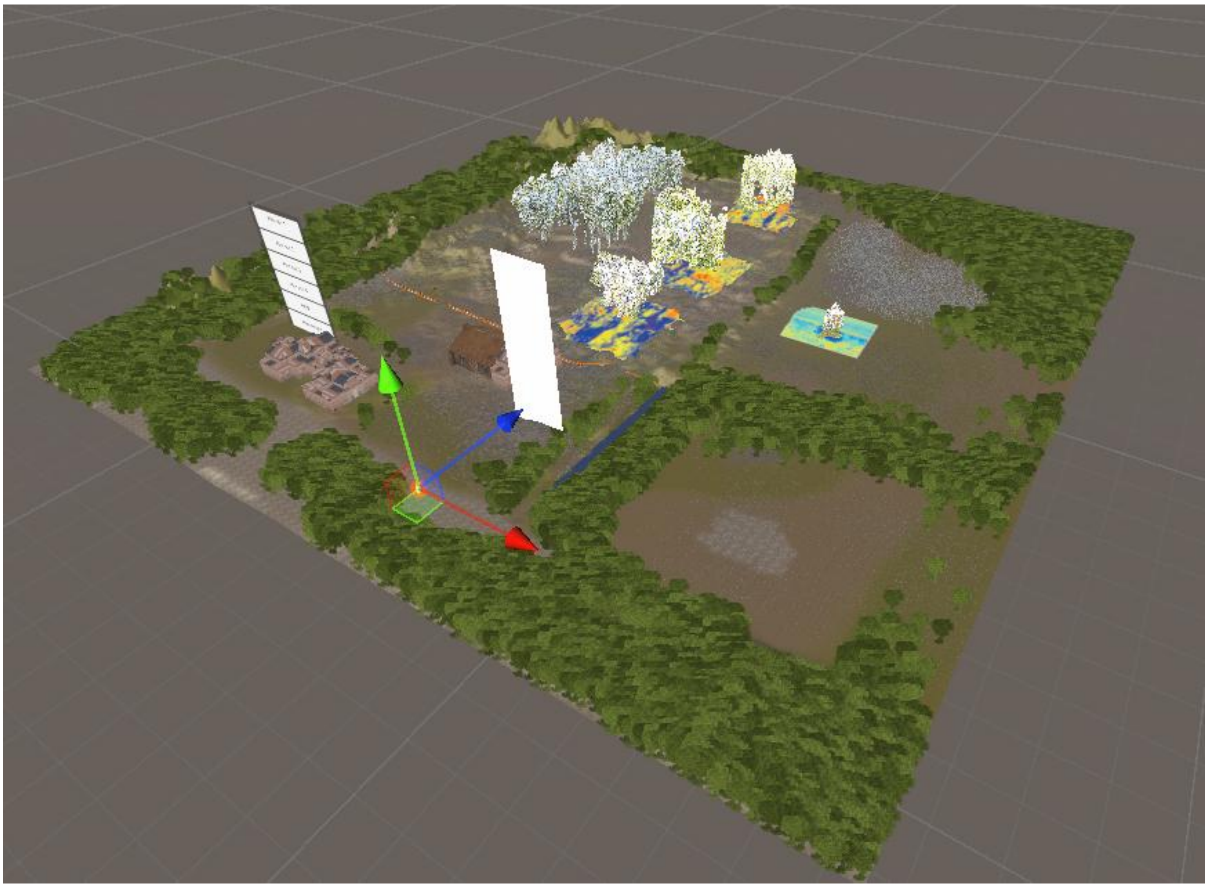


Figure 3: An overview of the scene in Unity. The LiDAR point cloud trees can be spotted in the middle.

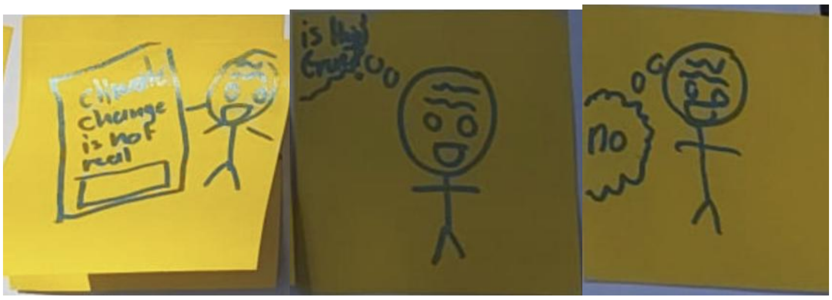
The figures show the final unity scenario. A live demo will be presented at the end of the assignment.

1.5. Justification

The purpose of our Digital twin of a forest is to get young people enthusiastic about the effect trees have on temperature. This will combine science education and real-world environmental awareness. We aim to make these somewhat abstract themes clear for younger generations. This way, the younger generation could get more excited about nature, and this could, in the end, cause them to tell their friends. This is also shown in the figure on the right, where a child enthusiastically talks to his peers about trees. This drawing was made for our wireframe during the last part of the excursion to Groenlo.

By using easy-to-understand terms, the children will be able to relate to it better and not be scared away by complicated terms and explanations. This overall should boost the experience and enthusiasm for most kids.

Also, by letting children be able to see the effect of temperature and vegetation themselves, they could think for themselves where this effect is coming from. This teaches them to critically evaluate situations and cause-and-effect relationships. This could teach them to think critically about all kinds of subjects, which is an important life skill. The goal of children to learn how to think critically is shown in the drawings, first the child encounters something, and questions if this is true, after that it comes to a conclusion on its own if the information is correct or not. These are the behaviours we want to stimulate by using our application. This drawing was also made when creating the wireframe at our excursion in Groenlo.



1.6. Conclusion

This project combines TLS and thermal data collected in Groenlo. This data was used to create a Unity game to teach children about the impact trees have on their surroundings. Our target audience already has some understanding of the impact of shade and cooling, but this application shows the actual impact vegetation has on its environment.

By using VR we make the experience more engaging for our target audience. This makes it easier for the target audience to capture and actually understand what's happening.

Since this is just a prototype there is a lot to improve. Future development could include more ecological information and adding more reliable data by going into the field more often collecting data during different weather conditions.

Overall we think this project is a foundation for making children be more aware and interested in nature.

2. 2. Data management plan

2.1. Organizational context

Name	Project group 3: CGI
Group members	Jurre Bakker, Niek Wels, Gabriel Castro Barrientos, Maarten Löhrer
Date	15/05/26
Chair group	Geo-Information Science and Remote Sensing (GRS)
GRS Supervisor	Kirsten de Beurs
Start date project	11/05/26
File name DMP	GRS3_HEAT_LIDAR_Data

2.2. Project description

Title	Exploring Nature's Cooling Effect Through a Digital Twin
Description	This project explores how local vegetation in the Groenlo forest affects temperature at ground level. Using a small drone, we collect temperature and LiDAR scan data above the forest. These are combined into an interactive 3D map, allowing students to visually compare plant density with local temperature differences.

2.3. Data management roles

Data collecting	Everybody collects data, some people work more with the LiDAR, whilst others work more with the drone.
Data analyzing	Similar to the collection; the work is split up, two people work on the heat data whilst the other two work with the LiDAR data.
Other contributions	Data validation, processing data
Role of supervisor	Support and advice

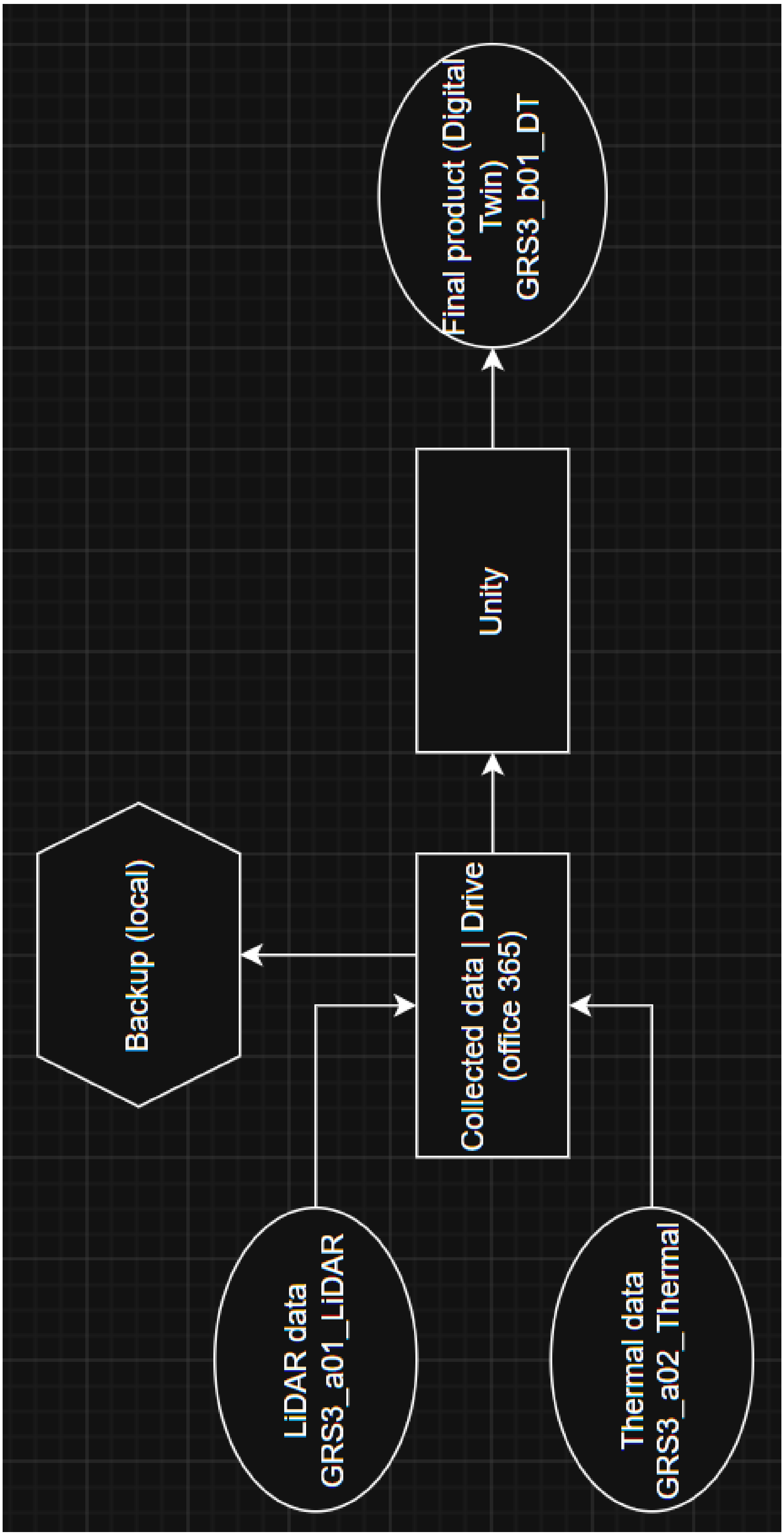
2.4. Project data

Data stage	**Specification of type of project data**	**Software/Hardware choice**	**Data size/ growth estimation**
Source data	RiCopter canopy LiDaR	Lidar point cloud	179 kb (zip)
Collected Data	LiDaR scan forests multiple locations	VZ-400	High GB (expectation)
	Thermal camera drone	Matrice drone with thermal attachment	Low MB (expectation)
Result data	3D space	Unity (version 2021.3.45F2)	10-20 GB (expectation)
Models/ Code	Digital twin	Unity (version 2021.3.45F2)	10-20 GB (expectation)

2.5. Short term storage

Data stage	**Storage location**	**Backup procedures**
Result data	Drive (Office 365 (WUR))	Local storage
Models/code	Unity	Redownload

2.6. Data and naming structure



2.7. Documentation and metadata

All files and products will be provided with proper and extensive metadata according to the ISO 19115:2013 standard. The final product will have a Readme with run and use description. File structures can be found in chapter 6.

2.8. Sharing and Ownership

Sharing and ownership	**With who/what/how**
Data sharing	Group members, teachers, experts
Data ownership	Measured data: Group members, teachers, experts Product: Group members
Privacy	The measured data will not be shared. The final product offers a view of forest that are accessible to the public.
Accesibility	Since this research serves for now more as a proof of concept the final product will only be accessible via personally contacting the group members or gaining access to the cloud teams environment. This can all be achieved by contacting any of the previously mentioned group members.
Accesibility final product	The final project can be requested by teachers or schools from the universities after which the schools can decide how to share this with their students

2.9. Long term storage

Yes or no?	**Argumentation**
Yes	The final product has to be stored long term for it to be used in educational activities. The storage will take place on the WUR provided teams/cloud environment where any WUR related party could theoretically gain access to it.

3. References

Allecijfers.nl (2026). Gemeente Oost Gelre. https://allecijfers.nl/gemeente/oost-gelre/, Data source was used on 13-05-2026

Statement on the use of AI:

1) ChatGPT 5.5 was used for the graphic logo by using the project description as a prompt.
2) ChatGPT 5.5 was used for a final spelling and grammar check